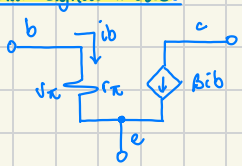
ELG 2136

lecture 7

AC analysis of BJT

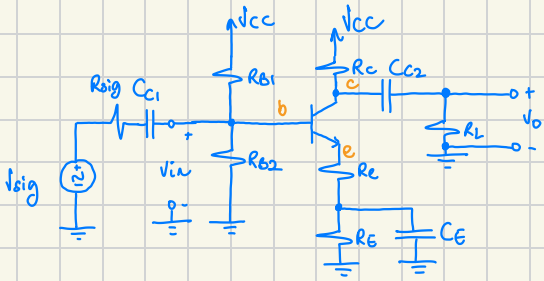
Small-signal model

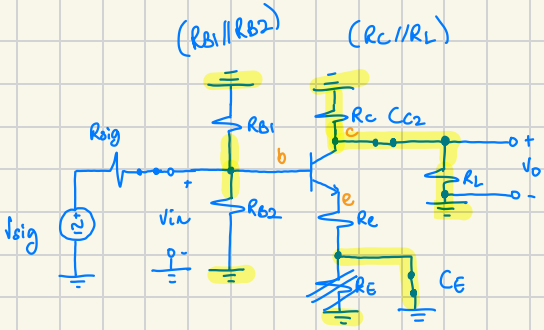


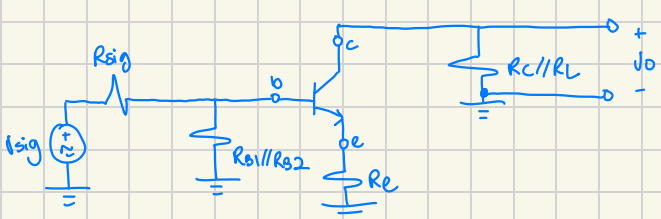
AC analysis:

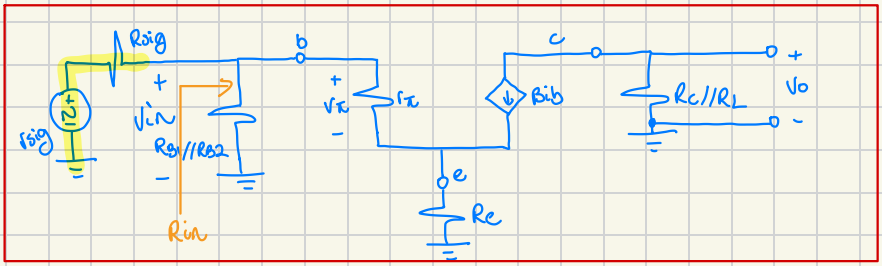
- Voltage gain: $A_v = v_o / v_i$
- Input resistance: R_{in}
- Overall voltage gain: $G_v = v_o / v_{sig}$
- Current gain: $A_i = i_o / i_i$
- Output resistance: R_{out}

- disable DC sources
- short circuit all capacitors
- replace BJT by small-signal model [Hybrid-π]

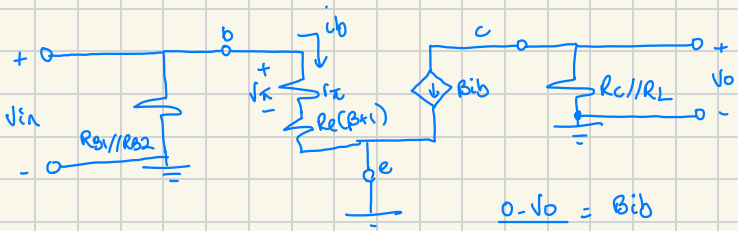








Transfer resistor rule: To bring a resistor from emitter to base, it must be multiplied by $(\beta+1)$

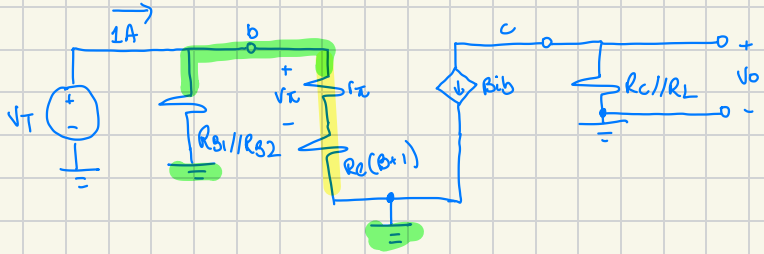


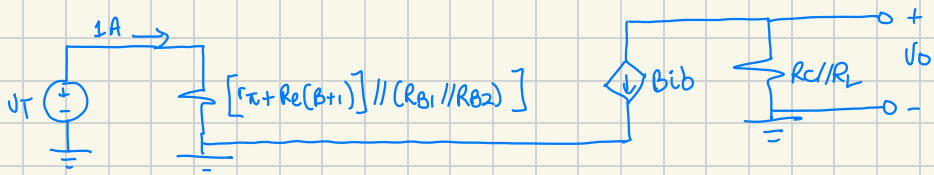
$$ib = \frac{Vin}{r\pi + (\beta+1)Re}$$

$$\frac{0 - Vo}{Rc//RL} = Bib$$

$$\Rightarrow \frac{-Vo}{Rc//RL} = \beta\left[\frac{Vin}{r\pi + Re(\beta+1)}\right]$$

$$\Rightarrow \boxed{Av = \frac{Vo}{Vin} = -\frac{\beta(Rc//RL)}{r\pi + Re(\beta+1)}}$$



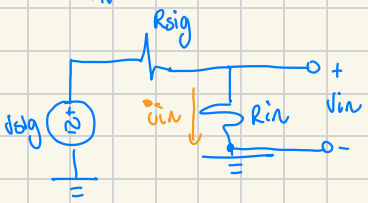


$$V_T = 1\left[r\pi + Re(\beta+1)\right] // (R_{S1} // R_{S2})\ ]$$

$$\boxed{R_{in} = \left[r\pi + Re(\beta+1)\right] // (R_{S1} // R_{S2})\ ]}$$

$$G_V = \frac{v_o}{v_{sig}} = \left(\frac{v_o}{v_i}\right)\left(\frac{v_i}{v_{sig}}\right) = A_V\left(\frac{v_i}{v_{sig}}\right)$$

A_V



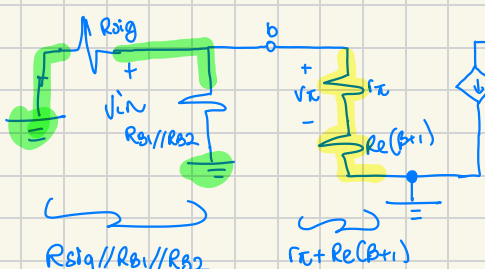
$$v_{in} = \left(\frac{R_{in}}{R_{sig}+R_{in}}\right) v_{sig}$$

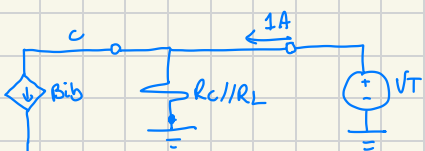
$$\Rightarrow \boxed{\frac{v_i}{v_{sig}} = \left(\frac{R_{in}}{R_{sig}+R_{in}}\right)}$$

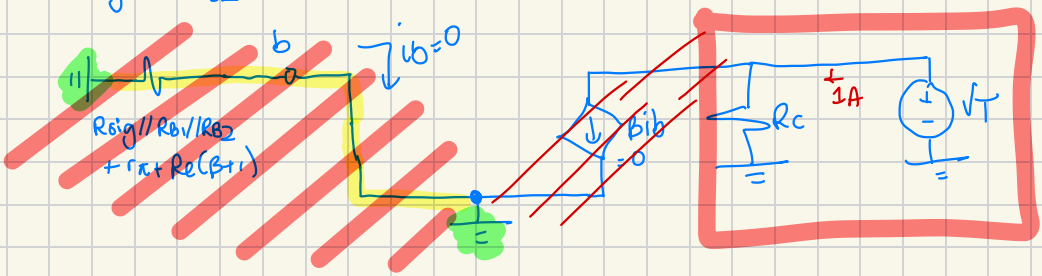
$$G_V = A_V\left[\frac{R_{in}}{R_{sig}+R_{in}}\right] = \left(\frac{-\beta(R_C /\!/ R_L)}{r_\pi + R_e(\beta+1)}\right)\left[\frac{[r_\pi + R_e(\beta+1)] /\!/ (R_{B1} /\!/ R_{B2})}{R_{sig} + [r_\pi + R_e(\beta+1)] /\!/ (R_{B1} /\!/ R_{B2})}\right]$$

$$A_i = \frac{i_o}{i_i} = \left(\frac{v_o}{R_L}\right)\left(\frac{R_{in}}{v_{in}}\right) = A_V\left(\frac{R_{in}}{R_L}\right) = \left[\frac{-\beta(R_C /\!/ R_L)}{r_\pi + R_e(\beta+1)}\right]\frac{[r_\pi + R_e(\beta+1)] /\!/ (R_{B1} /\!/ R_{B2})}{R_L}$$

$$i_{in} = \frac{v_{in}}{R_{in}} \qquad i_o = \frac{v_o}{R_L}$$







$$V_T = R_C$$

$$\Rightarrow \boxed{R_{out} = R_C}$$